

Your grade: 100%

Next item →

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

1. Which proof technique involves assuming the negation of what is to be proved and deriving a contradiction?

1 / 1 point

- ☒ Proof by contradiction
- ☐ Direct proof
- ☐ Proof by contraposition
- ☐ Proof by induction

👍 Nice work

Well done! Proof by contradiction involves assuming the negation of the statement and showing that this assumption leads to a contradiction.

2. What is the n th term of an arithmetic sequence where the first term is $a_1 = 3$ and the common difference is $d = 5$?

1 / 1 point

- ☐ $a_n = 3n + 5$
- ☐ $a_n = 3 + 5n$
- ☐ $a_n = 3 + n \times 5$
- ☒ $a_n = 3 + (n - 1) \times 5$

👍 Nice work

Well done! You've correctly applied the formula

$$a_n = a_1 + (n - 1)d$$

for the n th term of an arithmetic sequence.

3. What is the primary application of recursion in algorithms?

1 / 1 point

- ☒ To solve problems by breaking them down into smaller sub-problems
- ☐ To enhance the speed of mathematical calculations
- ☐ To optimize memory usage by reducing stack space
- ☐ To iterate quickly over large datasets

👍 Nice work

Well done! Recursion is used to break problems into smaller, more manageable pieces.

4. Which of the following best describes a recursively defined function?

1 / 1 point

- ☐ A function that relies on global variables to store intermediate results
- ☐ A function that processes data using iterative loops
- ☐ A function that operates independently without calling other functions
- ☒ A function that calls itself directly or indirectly to solve a problem

👍 Nice work

Great job! Recursively defined functions indeed call themselves to solve problems by breaking them down into smaller instances.

5. Which of the following expressions correctly represents the derivative of the function

1 / 1 point

$$f(x) = 3x^4 - 5x^3 + 2x - 7$$

?

- ☒ $f'(x) = 12x^3 - 15x^2 + 2$
- ☐ $f'(x) = 3x^3 - 5x^2 + 2$
- ☐ $f'(x) = 12x^3 - 15x^2 + 2x - 7$
- ☐ $f'(x) = 12x^4 - 15x^3 + 6x^2 - 7$

👍 Nice work

Nice work! Differentiation involves applying the power rule to each term of the polynomial function.